

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.9: Linear Motion — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.9: Linear Motion
Driving Question	DRIVING QUESTION: How can we use mathematics — calculations and graphs — to fully describe and explain the motion of any moving body, from a Nairobi Marathon runner to a matatu on the Nakuru highway?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Key Terms: What Is Really Happening in the Nairobi Marathon?	Watched the 2023 Nairobi Marathon video clip showing Kenyan elite runners accelerating off the start on Uhuru Highway, running at a steady pace, slowing at water stations, and sprinting to the finish. Examined the split-time table for Runner A (consistent pace) and Runner B (fast start, mid-race slowdown, strong sprint) and noticed that Runner B was ahead at 30 km but Runner A won. Conducted a class race on the school field, recording our own split times at 10 m intervals using stopwatches and tape measures.	Distance is the total path length covered (a scalar quantity — no direction needed). Displacement is the straight-line change in position from start to finish, including direction (a vector quantity). Speed = distance ÷ time (scalar). Velocity = displacement ÷ time with direction (vector). Acceleration is the rate of change of velocity — it explains why Runner B's split times changed dramatically between 30 km and 42 km. We calculated Runner A's average speed as approximately 19 km/h, and we calculated our own average speed over each 10 m interval of the class race.	Runner A won the 2023 Nairobi Marathon even though Runner B was ahead at 30 km because Runner B's velocity decreased sharply in the middle kilometres while Runner A maintained a nearly constant velocity throughout. Runner B's sprint in the final 2 km was not enough to overcome the time lost. A single average speed calculation cannot reveal this — we need a graph that shows what is happening at every moment of the race. Our class race data table is our first motion model, and in the next lesson we will convert it into a displacement-time graph to see the full story of motion.
Lesson 2 Calculating Velocity and Acceleration — Making the Numbers Tell the Story	Learners generated a velocity table for Runner A and Runner B by applying $v = s/t$ to each 5 km split of the 2023 Nairobi Marathon data. They observed that Runner A maintained a nearly constant velocity (~19 km/h) throughout — uniform motion — while Runner B's velocity varied from ~17.6 km/h in the 25–35 km section to ~22 km/h in the final sprint — non-uniform motion. They also calculated the acceleration of a	Velocity is calculated as $v = s/t$ (displacement divided by time) and is measured in m/s or km/h. Acceleration is calculated as $a = \Delta v/\Delta t$ (change in velocity divided by time taken) and is measured in m/s^2 . Uniform motion means constant velocity and zero acceleration; non-uniform motion means changing velocity and non-zero acceleration. A positive acceleration means the object is speeding up; a negative	The split-time data from the Nairobi Marathon phenomenon can now be expressed as a sequence of velocities and accelerations — not just a single average speed. Runner B was ahead at 30 km because his early high velocity (positive acceleration off the start) built a time advantage. However, his large negative acceleration (deceleration) between 25 km and 35 km cost him more time than his final

	boda-boda on Mombasa Road (1 m/s^2) and the SGR Madaraka Express, and used $a = \Delta v/\Delta t$ to find Runner B's acceleration between the slow and fast sections of the race.	acceleration (deceleration) means it is slowing down. Runner A's near-constant velocity explains his consistent split times; Runner B's large positive and negative accelerations explain why his split times varied so dramatically.	sprint could recover. Runner A's near-zero acceleration — uniform motion — meant every segment was equally efficient, and this consistency ultimately won the race. The mathematical tools $v = s/t$ and $a = \Delta v/\Delta t$ reveal the internal structure of motion that a single total-time comparison completely hides.
Lesson 3 Drawing Displacement-Time Graphs: Reading the Shape of the Race	Learners plotted the split-time data for Runner A and Runner B from the 2023 Nairobi Marathon on the same displacement-time graph using squared paper, two colours, labelled axes, and a title. They circled the point where Runner B's line crossed Runner A's line — the graphical moment that explains the race outcome.	The shape of a displacement-time graph — not just its start and end points — tells the full story of motion. A steep line means high speed; a gentle line means low speed; a horizontal line means the runner is stationary. The gradient of the line at any section represents the speed of the runner at that stage of the race.	Using the completed dual displacement-time graph, learners explained why Runner B — despite being ahead at 30 km — lost the race: Runner B's line became noticeably less steep between 25 km and 35 km (slowing down), while Runner A's line maintained a consistent gradient throughout, overtaking Runner B during the slow section between 30 km and 35 km. A single average-speed calculation could never have revealed this — only the graph shape can.
Lesson 4 Interpreting Displacement-Time Graphs: Gradient as Velocity	Learners examined four pre-drawn displacement-time graphs showing Kenyan real-life motion scenarios: a matatu stopping at a Nakuru stage, a marathon runner resting at a water station on Uhuru Highway, a boda-boda rider returning home, and the SGR Madaraka Express making stops between Nairobi and Mombasa. Using the Gradient Hunter table, learners read coordinates from each segment and calculated gradient values for every phase of each journey.	Learners established that the gradient of a displacement-time graph equals velocity; a zero gradient means the object is stationary; a negative gradient means the object is returning toward its starting point; and a steeper positive gradient means higher velocity. Learners applied this to the Marathon phenomenon by annotating Runner A and Runner B's sketch graphs with segment gradient values and identifying that Runner A's higher gradient while Runner B slowed (25–35 km) is the mathematical explanation for why Runner A won despite Runner B being ahead at 30 km.	Learners explained that displacement-time graphs are used in real life to reveal the velocity at every phase of a journey — not just the overall average. By calculating the gradient of each segment, we can identify when a body was moving fast, slow, stationary, or returning. This answers Key Inquiry Question 2: we use displacement-time graphs to describe and compare motion in detail, segment by segment, which a single average speed calculation cannot do.
Lesson 5 Drawing Velocity-Time Graphs — Reading Acceleration in the Shape of the Race	Learners observed that a velocity-time graph for a matatu journey on Uhuru Highway, Nairobi, produces three visually distinct segments: a rising straight line (acceleration from the Kencom bus stop), a horizontal straight line (constant cruising velocity), and a falling straight line (braking before the next stage). They also observed Runner B's velocity-time sketch derived from marathon split data, showing a mid-	The gradient of a velocity-time graph equals acceleration ($a = \Delta v/\Delta t$, measured in m/s^2). A horizontal segment means zero acceleration (constant velocity); a positive gradient means the object is speeding up; a negative gradient means it is decelerating. The shape of the velocity-time graph — its slopes, not just its start and end points — quantifies how rapidly velocity is changing at every moment of the journey.	The velocity-time graph of the Nairobi Marathon now mathematically explains what the split-time tables could only hint at: Runner B's steep positive gradient between km 35 and km 42 confirms a genuine sprint-finish acceleration, but the graph also shows that his velocity during km 25–35 was significantly lower than Runner A's flat, consistent velocity profile across the same period. The gradient calculations make it

	race dip in velocity followed by a steep positive gradient in the final 2 km sprint.		possible to state, with numbers and units, exactly how much Runner B accelerated — and why, despite that acceleration, his overall average velocity was lower, giving Runner A the victory.
Lesson 6 Interpreting Velocity-Time Graphs: Area Under the Graph as Distance	Learners plotted and examined velocity-time graphs for two Kenyan real-world contexts: the SGR Madaraka Express journey out of Nairobi and a 400 m athletics race at Nakuru High School. They identified and labelled three motion phases — acceleration, constant velocity, and deceleration — by observing the shape and slope of each graph segment. They also observed a simplified velocity-time graph comparing Runner A and Runner B's motion between km 25 and 35 of the 2023 Nairobi Marathon.	Learners learned that: (i) the gradient of a velocity-time graph equals acceleration — a steep positive gradient means rapid acceleration, a negative gradient means deceleration, and a zero gradient (horizontal line) means constant velocity; (ii) the area of the region between the velocity-time graph and the time axis equals the total distance (displacement) travelled; (iii) this area can be calculated by decomposing the region into rectangles and trapeziums, or by counting grid squares; (iv) comparing the areas under Runner A's and Runner B's velocity-time graphs for km 25–35 mathematically proves that Runner A covered more ground in that segment and therefore overtook Runner B to win the race.	Learners explained, using their area calculations, why Runner B lost the 2023 Nairobi Marathon despite leading at the 30 km mark: Runner B's velocity dropped significantly between km 25 and 35 (smaller area under the graph = less distance covered per unit time), while Runner A maintained a higher velocity for longer (larger area = more distance covered). This resolved the central puzzle of the phenomenon and directly answered the key inquiry question 'How do we use velocity-time graphs?' Learners updated their cumulative models and the Driving Question Board to reflect this mathematical explanation.
Lesson 7 Relative Speed — Who Gets There First When Everyone Is Moving?	We observed (through calculation) that a matatu travelling at 90 km/h approaches a bus travelling in the same direction at 60 km/h at a closing speed of only 30 km/h — meaning the overtake takes much longer than the driver expects. We also observed that two Lake Victoria ferries travelling towards each other close the gap between them at 40 km/h (the SUM of their speeds), and that Runner A appeared to approach Runner B at just 1.4 km/h when overtaking — the DIFFERENCE of their velocities.	We learned that RELATIVE SPEED is the speed of one body as observed from another moving body. When two bodies move in the SAME direction, relative speed = faster speed – slower speed (they approach each other slowly). When two bodies move in OPPOSITE directions, relative speed = speed A + speed B (they approach each other rapidly). We can use relative speed to calculate the time for two bodies to meet or for one to overtake the other, using the formula: $\text{time} = \frac{\text{gap distance}}{\text{relative speed}}$.	This explains a crucial part of the marathon phenomenon: Runner A did not need to run faster than Runner B's earlier pace to overtake him — Runner A only needed to run faster than Runner B's CURRENT speed. Because Runner B had slowed to about 17.6 km/h and Runner A was running at about 19 km/h, the relative speed between them was about 1.4 km/h. At that closing rate, Runner A covered the 200 m gap in approximately 8.6 minutes. The same mathematics also explains the NTSA road-safety context: a head-on collision between two vehicles both doing 80 km/h has a relative speed of 160 km/h — four times more dangerous than a rear-end collision between a 90 km/h matatu and a 60 km/h bus. Mathematics does not just describe motion — it saves lives.
Lesson 8 Consolidation, Model Building & Final	Learners re-watched the 2023 Nairobi Marathon video clip showing elite Kenyan runners — including Runner A and Runner	Learners produced a complete mathematical explanation of the marathon race, calculating average speed, segment	Runner A won the 2023 Nairobi Marathon because of consistently high velocity maintained throughout the race, visible as a

Explanation: The Full Story of the Nairobi Marathon	B — with different pacing strategies. They examined the printed split-time table showing each runner's time at every 5 km mark from 0 to 42 km, observing that Runner B led at 30 km but Runner A finished first. Groups plotted dual displacement-time and velocity-time graphs using the split-time data, and posted their A3 graphs on the classroom wall for a gallery walk comparison.	velocity, and acceleration; drawing and interpreting displacement-time graphs (where gradient equals velocity) and velocity-time graphs (where gradient equals acceleration and area equals displacement); distinguishing between average speed and instantaneous velocity; and applying relative speed concepts. They closed all three KICD key inquiry questions on the Driving Question Board with evidence-based answers drawn directly from the marathon data, and completed an individual written assessment covering all Sub-Strand 2.9 learning outcomes.	steep, nearly constant gradient on the displacement-time graph. Runner B's velocity decreased sharply between 25 km and 35 km — shown by a shallower gradient on the displacement-time graph and a negative slope (deceleration) on the velocity-time graph — creating a time deficit that Runner B's final sprint (positive acceleration in the last 2 km) could not overcome. This demonstrates that in linear motion, the shape of the graph — not just the start and end points — tells the full mathematical story of motion.
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